Using Particle Velocimetry to Investigate Energy Dissipation in Turbulent Eddies

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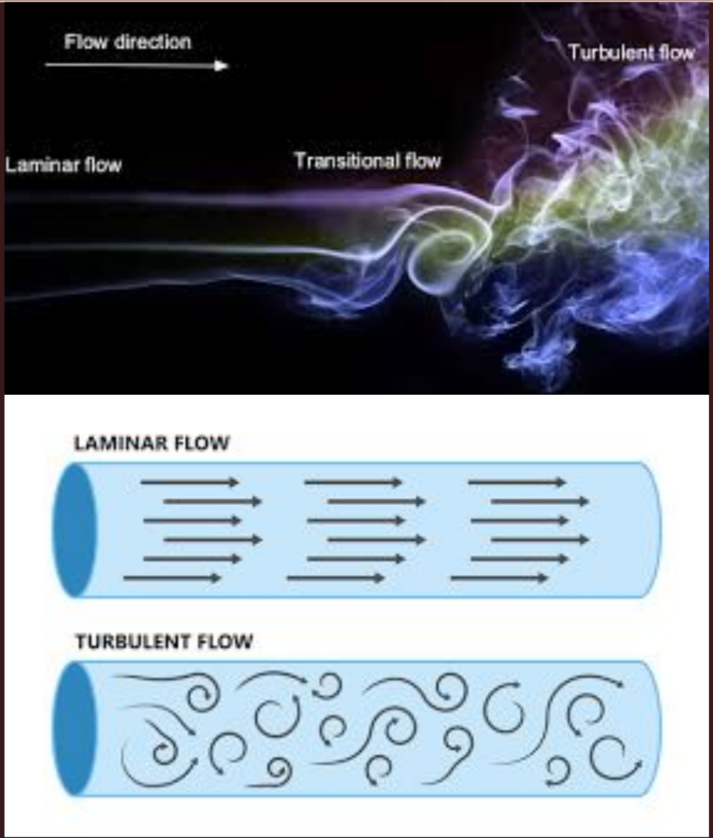
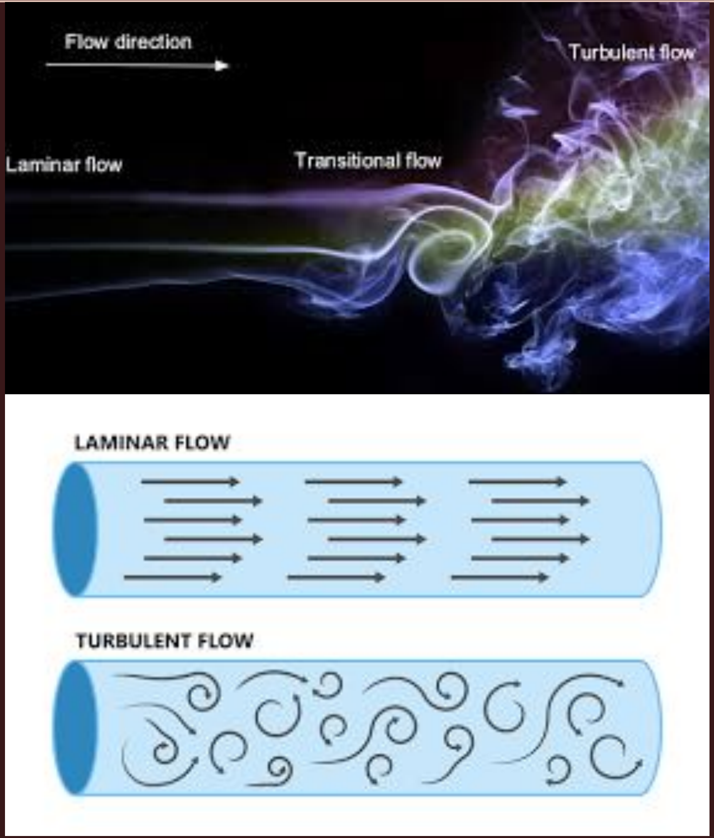
Introduction

- "Turbulence is one of the most important unsolved problems in physics"
- Has vortexes, eddies, and frequent changes in velocity



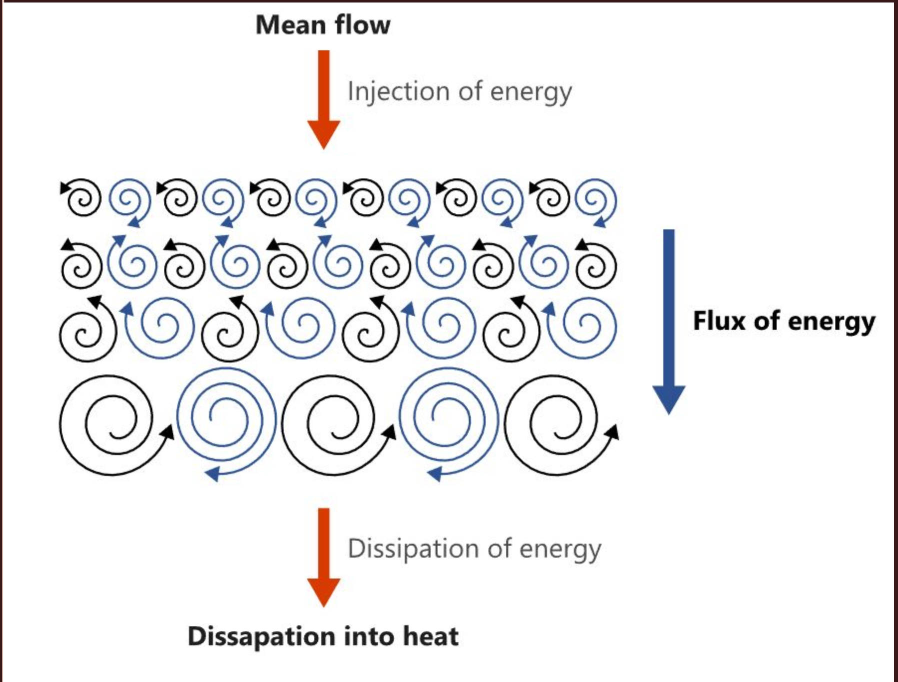
Solving Turbulence Would Mean…

- A drastic increase in energy efficiency for planes
- Billions of dollars saved by more efficient wind turbines
- Better medicine delivery
- Fields Medal



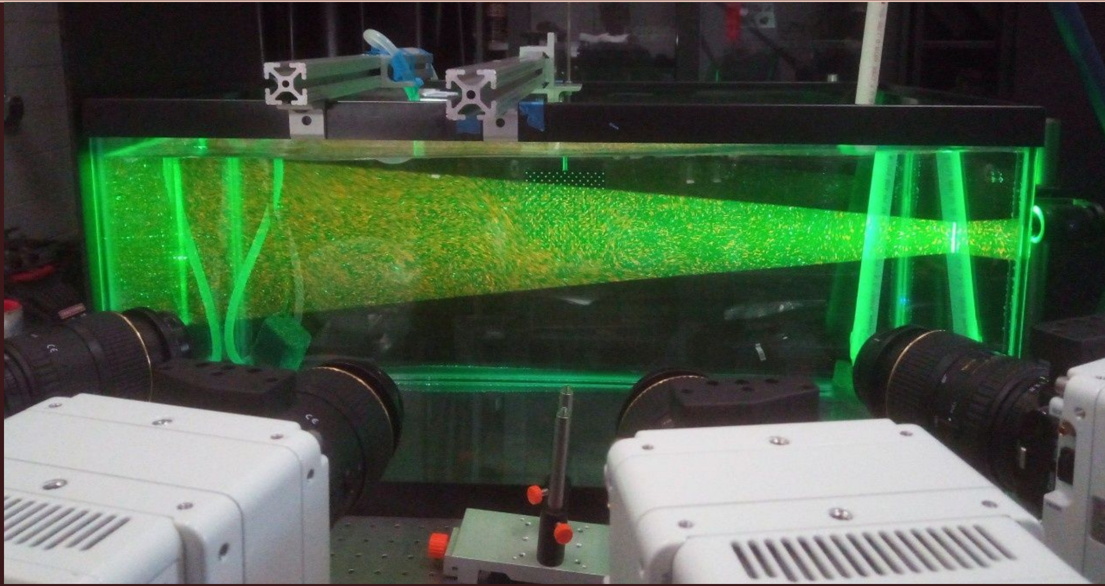
The Energy Cascade

- Transferred from kinetic energy of flow to molecular rotation (heat)
- Between different eddy sizes
- Scientists don't know *how*
- In 2D, energy cascades from small to large scales to heat



Particle Image Velocimetry(PIV)

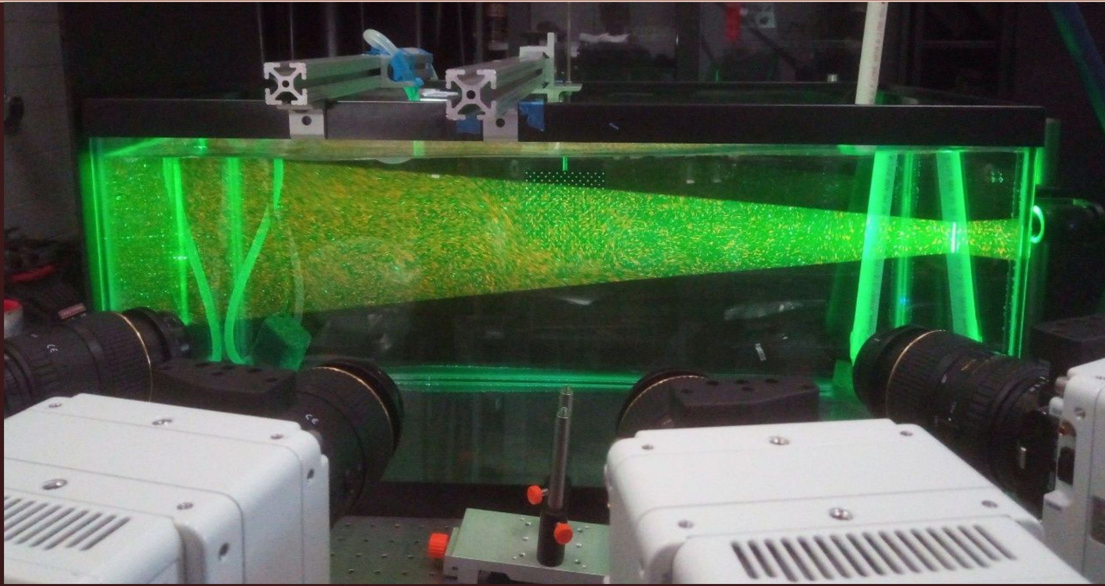
- Noninvasive method to find velocity fields of fluids
- Goal is to find



$$\mathbf{v}(\mathbf{r}, t) \quad \text{where } \mathbf{r} = (x, y, z)$$

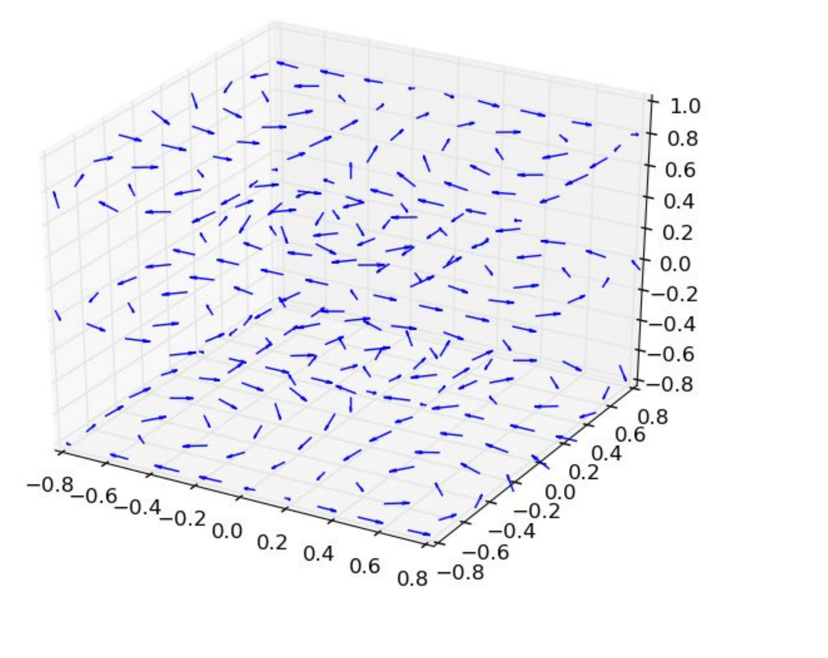
Particle Image Velocimetry(PIV)

- Tracer particles are assumed to exactly follow the flow
- Ideally semi buoyant



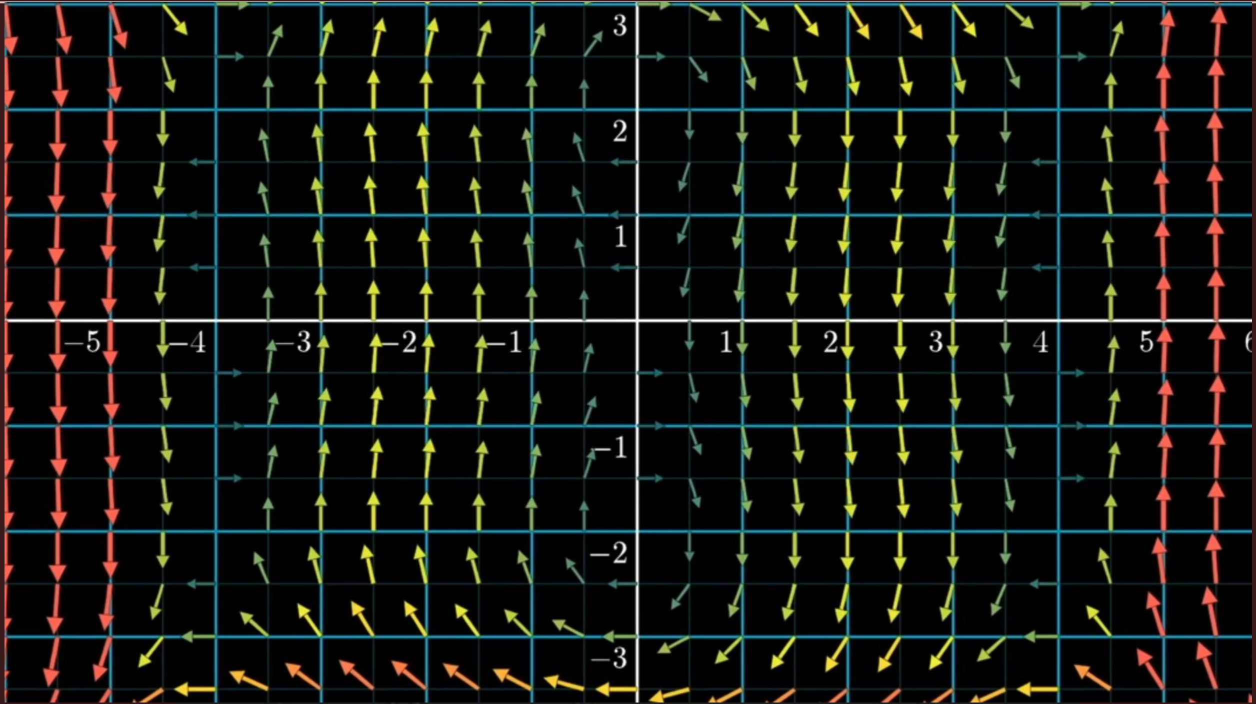
$$\mathbf{v}(\mathbf{r}, t) \quad \text{where } \mathbf{r} = (x, y, z)$$

3D Vector Field



$$\mathbf{v}(\mathbf{r}, t) \quad \text{where } \mathbf{r} = (x, y, z)$$

2D Vector Field



$$\mathbf{v}(\mathbf{r}, t) \quad \text{where } \mathbf{r} = (x, y)$$

Kinetic Energy of a Fluid:

$$K = \frac{1}{2}\rho h \iint_{\Omega} u^2 \, \mathrm{d}A$$

$$\boxed{K \approx \frac{1}{2}\rho h \sum_{i,j} \left(u_{x,ij}^2 + u_{y,ij}^2\right) \Delta A}$$

$$K = \frac{1}{2}mv^2$$

$$\mathrm{d}m = \rho \, \mathrm{dV}$$

$$\mathrm{d}K = \frac{1}{2}v^2 \, \mathrm{d}m$$

$$\mathrm{dV} = h \, \mathrm{d}A$$

$$\mathrm{d}K = \frac{1}{2}\rho v^2 \, \mathrm{dV}$$

Research Question:

Will there be a correlation between decay of the kinetic energy of a turbulent 2D flow and time?

My Experiment:

- Preform 2D PIV analysis on the surface of a shallow dish
- Use sesame seeds as tracers
- Find a **relationship between the kinetic energy of the system and time**

My Experiment:

- Controlled:
  Depth of water
  Speed/time of mixing
  Location of mixing

Procedure(Experimentation):

1. Set up the tray with black fabric covering the bottom.
2. Fill with water to one inch.
3. Insert around 50 sesame seeds
4. Put 1-2 drops of dish soap on the water.
5. Set up a black umbrella above the dish such that it blocks all reflection from the ceiling.

6. Set up a camera such that the plane of view is completely aligned with the surface of the water. Start recording.

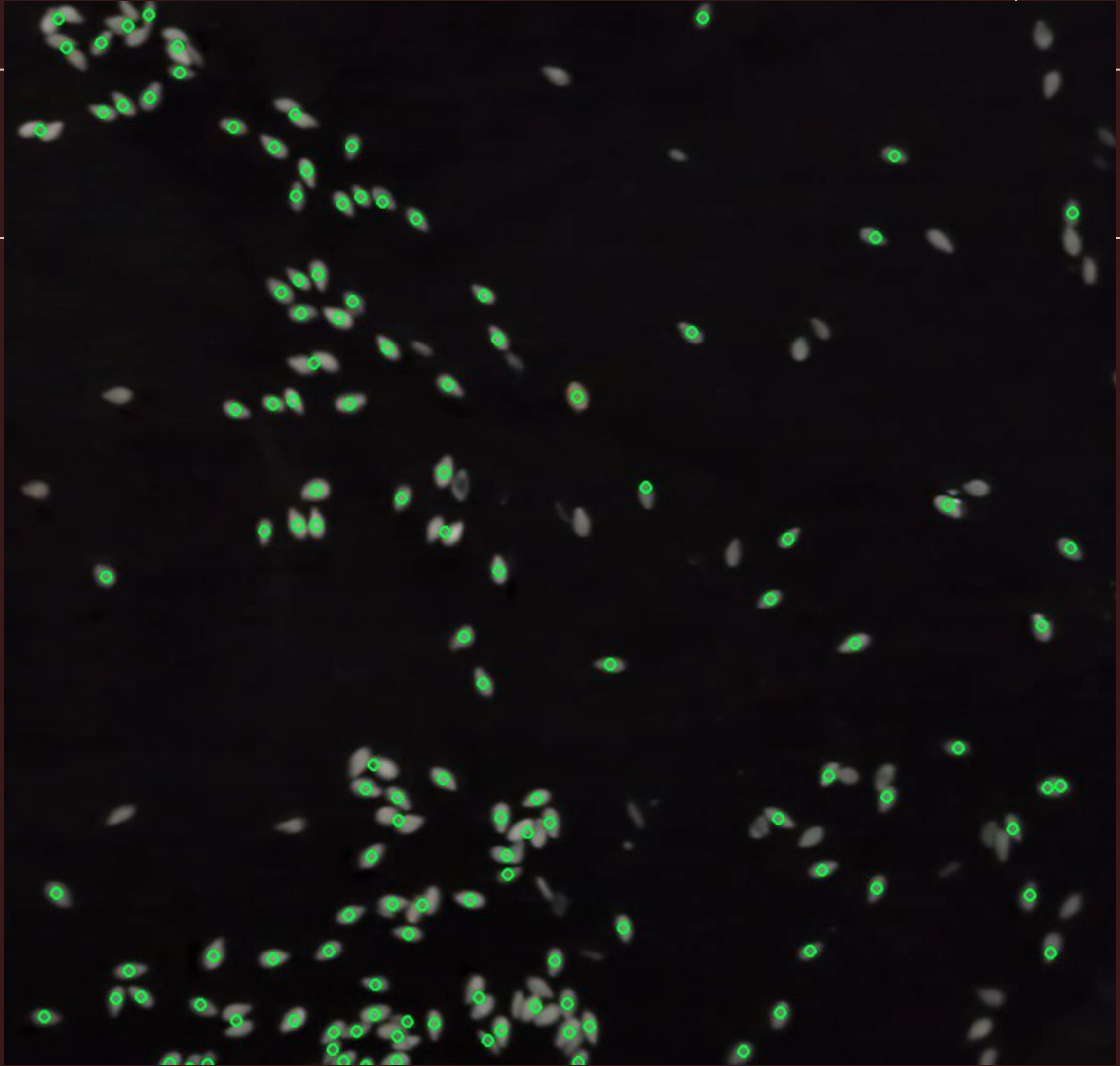
7. Spin the blender at the midpoint of the lower half of the dish for *t* seconds, making sure the blade is submerged. Repeat in the upper half immediately.

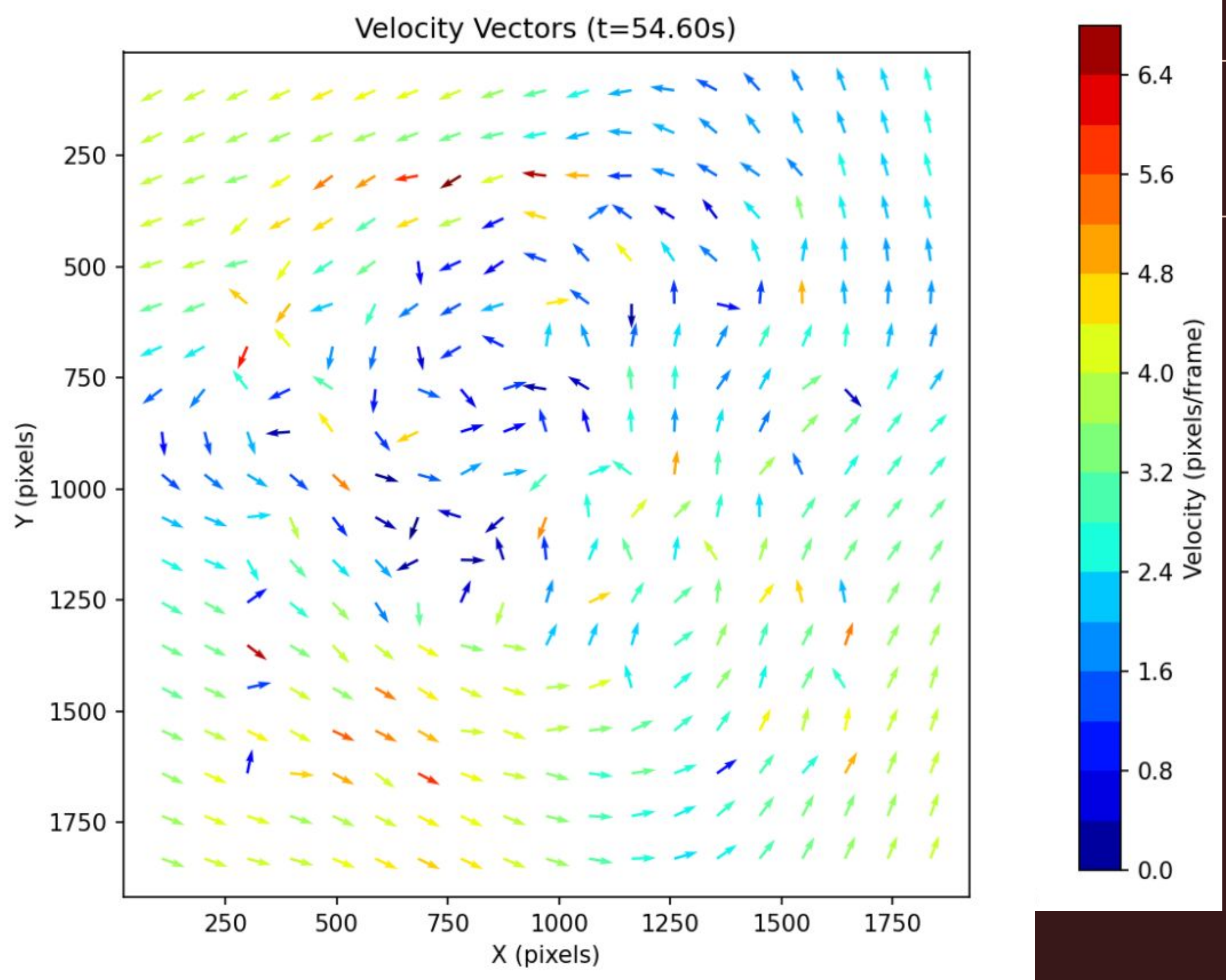
Procedure(Data Analysis):

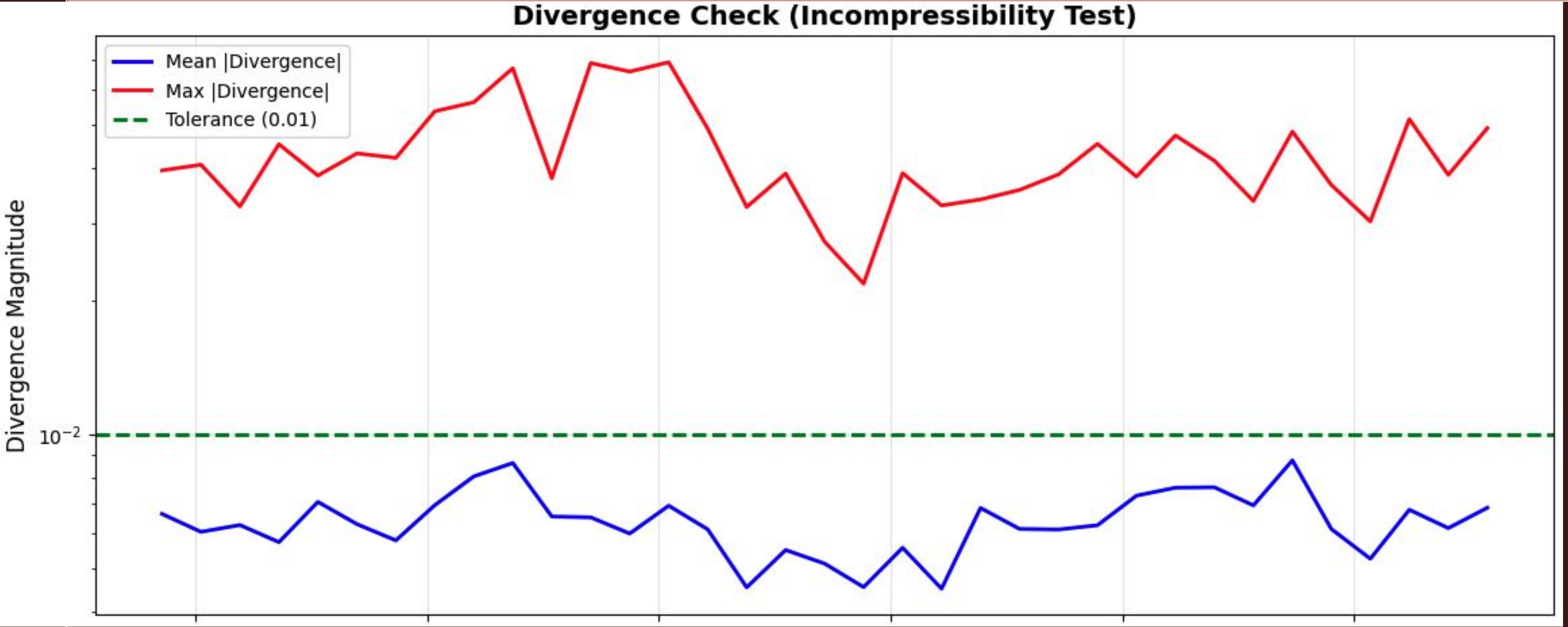
1. Use PIV to extract velocity fields from the duration of the video.
   - Take out outlier vectors
   - Force divergence

2. Run the discretized energy integral on each vector field.

3. Do an exponential regression of the energy vs time.

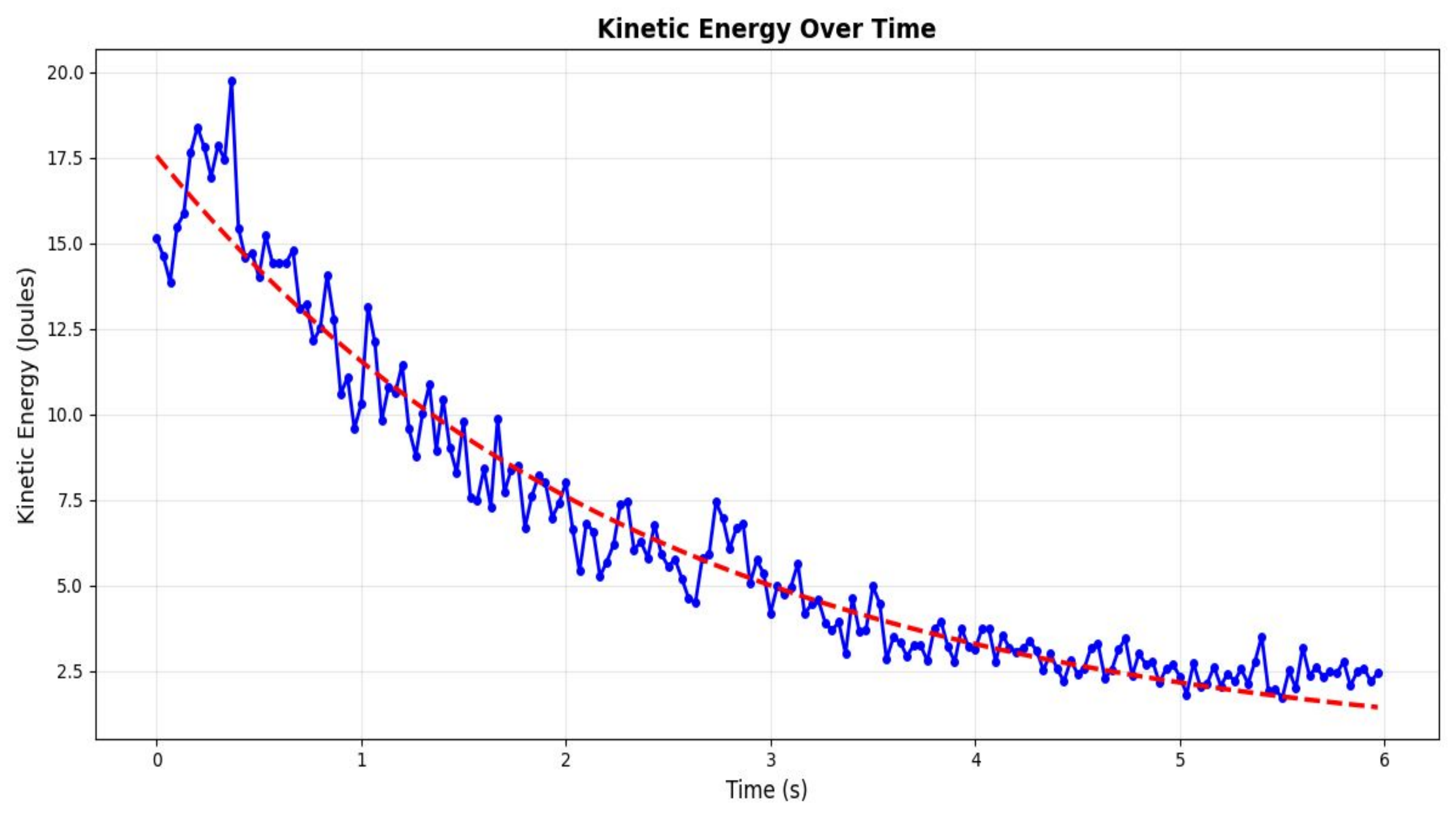



Velocity Vectors (t=54.60s)
Y (pixels)
X (pixels)
250
500
750
1000
1250
1500
1750
Velocity (pixels/frame)
6.4
5.6
4.8
4.0
3.2
2.4
1.6
0.8
0.0


Divergence Check (Incompressibility Test)
Mean |Divergence|
Max |Divergence|
Tolerance (0.01)
Divergence Magnitude
10^{-2}

Results: 2 Seconds Stirring

$$E(t) = E_0\, e^{-t/\tau}$$

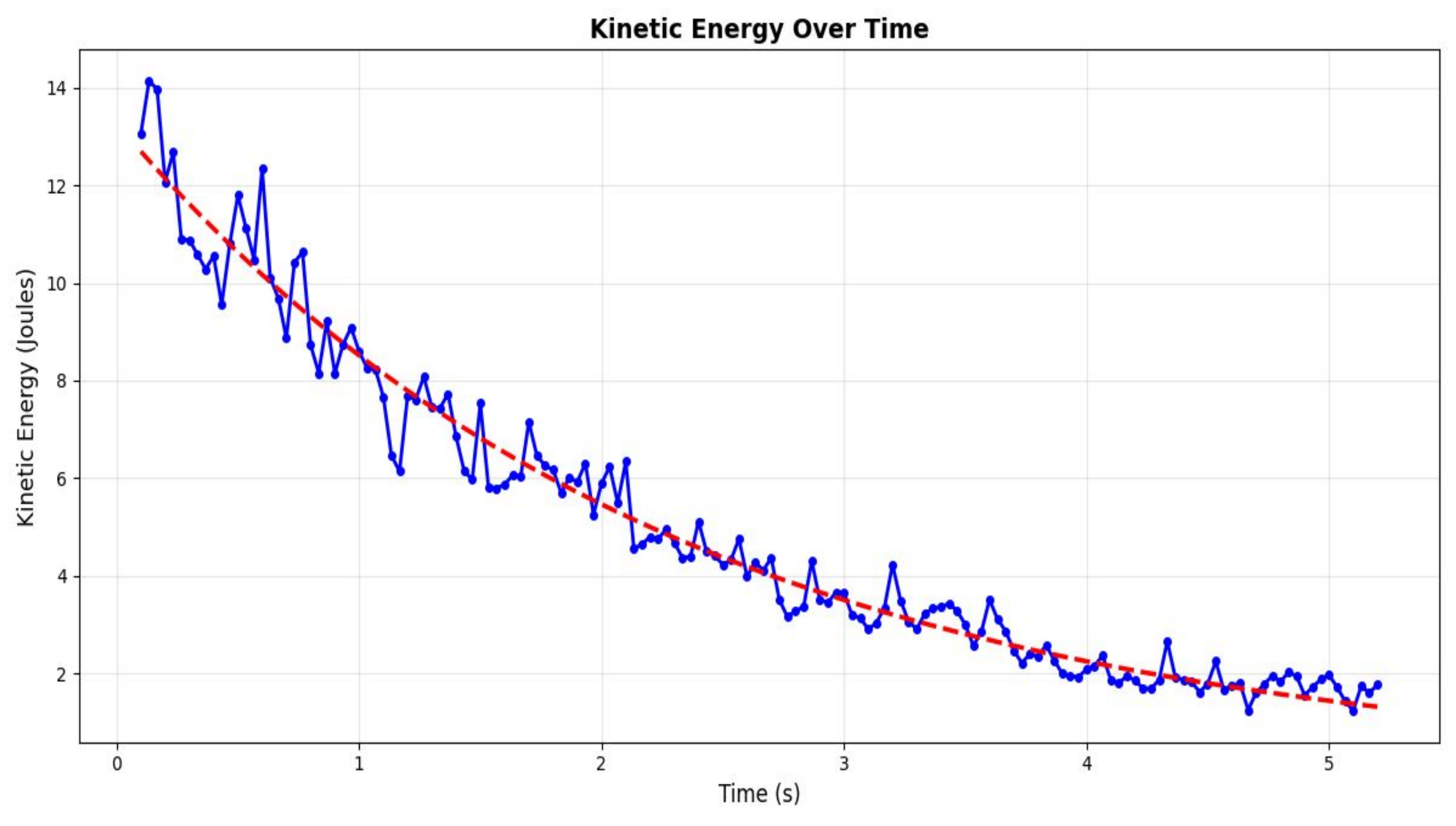


$R^2 = 0.95$

T = 2.39

Results: 5 Seconds Stirring

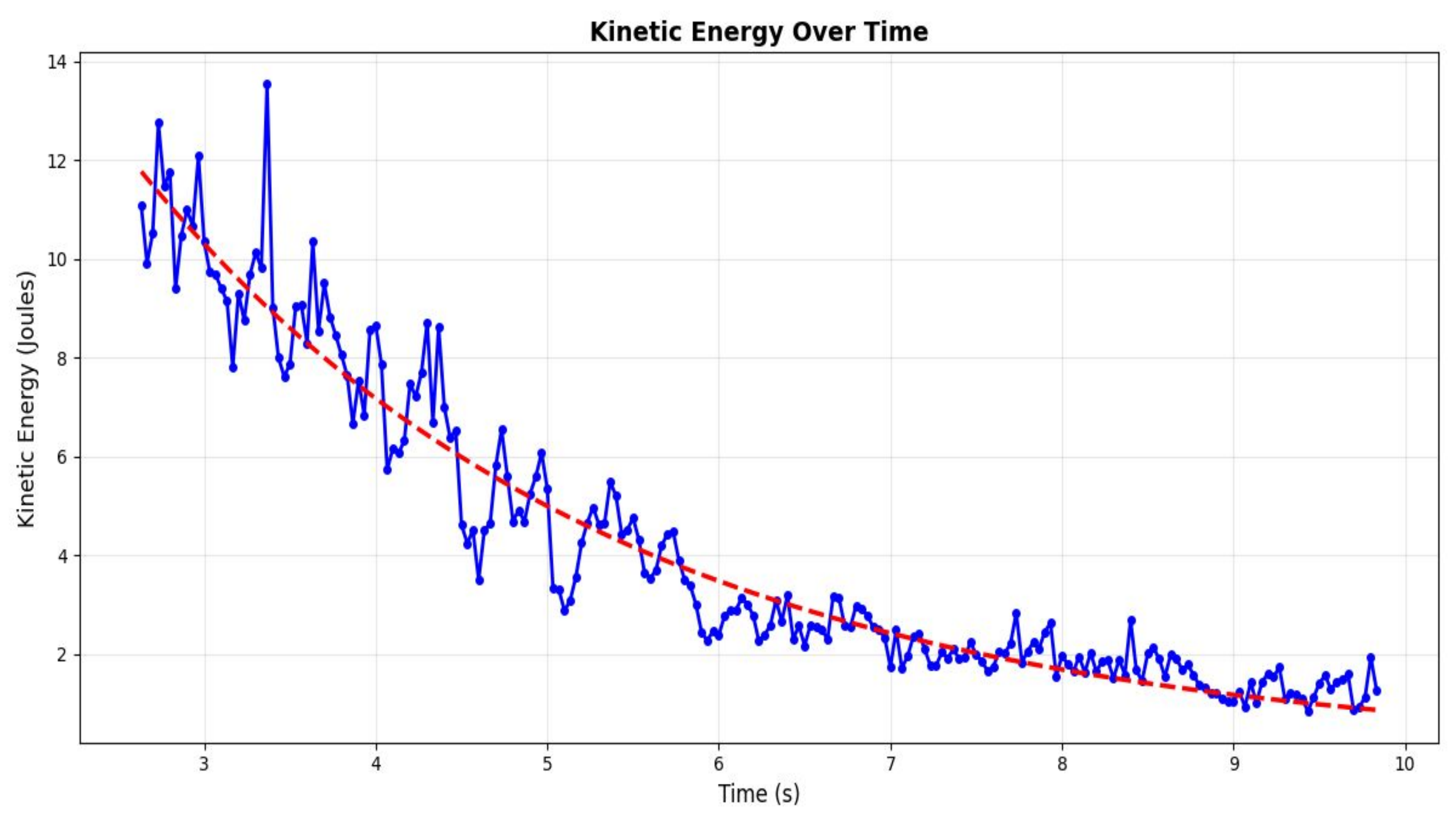
$$E(t) = E_0\, e^{-t/\tau}$$



R^2 = 0.97
T = 2.26

Results: 10 Seconds Stirring

$$E(t) = E_0\, e^{-t/\tau}$$



R^2 = 0.93
T = 2.76

Results:

- The plot showed a very clear, exponential relationship
- This reveals that although there is complexity, there is very clear order in the system
- Decays to heat at an exponential rate, can be reliably predicted

Limitations and Areas of Further Research:

- Blank areas formed in the wake of the mixer
- Darkness on the video
- Investigate the time constant
- Do it with different dish sizes

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